

SUMMARY

M.Sc. Scientific Computing at Heidelberg University & Working Student at NEC Labs Europe. Building production LLM systems, hybrid RAG pipelines, and multi-agent architectures. Strong mathematical foundation with hands-on experience shipping structured generation, retrieval, and evaluation infrastructure.

TECHNICAL SKILLS

Languages: Python, SQL, Bash, JavaScript

AI/LLM Systems: RAG (hybrid retrieval, reranking), LLM evaluation, multi-agent orchestration, LangGraph, Google ADK

ML/DL: PyTorch, Transformers, Sentence-Transformers, scikit-learn, ONNX Runtime, FlashRank

Infrastructure: FastAPI, Qdrant, ChromaDB, SQLite, BM25, Pydantic, Docker, Git, Linux, pytest

EXPERIENCE

Working Student – Software Development, NEC Laboratories Europe

May 2025 - Present

- Built internal LLM-powered prototypes for document intelligence, structured generation, and retrieval-augmented workflows serving research teams.
- Engineered multi-source ingestion pipelines (PDFs, URLs, text) with chunking, hybrid retrieval, source attribution, and structured output generation.
- Designed schema-constrained generation pipelines with JSON/Pydantic validation, retry logic, and automatic repair, reducing malformed output rates.
- Developed synthetic longitudinal data-generation workflows with cross-artifact consistency validation, stateful regeneration, RAG-backed reports, and token/cost tracking.

AI Data Specialist – LLM Evaluation, Soul AI

Oct 2023 - Apr 2024

- Contributed to RLHF data pipelines evaluating LLM outputs on college-level mathematics, targeting multi-turn reasoning, hallucination reduction, and context-retention failures.
- Reviewed and annotated 500+ evaluation tasks, identifying systematic failure patterns and collaborating with human evaluators to improve model quality.

SELECTED PROJECTS (PORTFOLIO)

EU Financial Regulation Hybrid RAG System (*Python, FastAPI, Azure OpenAI, Qdrant, BM25, FlashRank, ONNX*)

- Built a hybrid RAG system over 382 parsed articles from 4 EU regulations (PSD2, MiFID II, GDPR, DORA), combining BM25 lexical search with Qdrant dense retrieval fused via Reciprocal Rank Fusion.
- Implemented a query refinement layer using LLM-based intent classification (lookup / conceptual / compound) with automatic query decomposition, achieving 100% Precision and Recall at k=5.
- Integrated FlashRank ONNX reranking (34 MB model, ~100 ms latency on CPU) as a lightweight alternative to PyTorch Cross-Encoders, reducing context tokens by up to 35%.

Multi-Agent Research Assistant (*Python, Google ADK, FastAPI, ArXiv/PubMed/OpenAlex APIs*)

- Designed a multi-agent system using Google ADK with 7 specialized agents that search academic databases (ArXiv, PubMed, OpenAlex), cluster papers, and synthesize structured literature reviews.
- Implemented evidence quality assessment and automated citation extraction, producing publication-ready research summaries from natural-language queries.

SELECTED RESEARCH PROJECTS

Early-Exit Inference for BERT with Exit-Aware Verification ([GitHub](#))

- Implemented entropy, margin, and patience-based early-exit strategies with micro self-verification; measured true halting latency via layer-wise execution and CUDA timing across classification tasks.

Character-Aware Encoder Robustness Under Typos ([GitHub](#))

- Benchmarked CANINE vs. SBERT under controlled synthetic typo noise; evaluated embedding stability via Retrieval@1 and cosine-similarity metrics across multiple corpora.

EDUCATION

2024 - Present Master's Degree in Scientific Computing at **Universität Heidelberg, Germany**

2020 - 2023 Bachelor's Degree in Mathematics at **Fergusson College, Pune, India**